THE RESOLUTION PROPERTY DESCENDS ALONG A MORPHISM WITH AN F-AMPLE INVERTIBLE MODULE

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Allowed standard prerequisites (editorial).

- On the inverse image of an affine open of Y the f-ample invertible module behaves like an ample invertible module, so every finite type quasi-coherent module admits a jointly surjective finite family of maps from tensor powers of it. invoked at proof lines 9477-9483. Produces the finite families of sections over each piece of the cover.
- Every section of a quasi-coherent module over a quasi-compact quasi-separated scheme is contained in a finite type quasi-coherent submodule. invoked at proof lines 9492-9496. Trims the pushforward modules to finite type submodules so the resolution property of Y applies.
- The resolution property of Y produces surjections from finite locally free modules onto finite type quasi-coherent modules. used at proof lines 9497-9504. Supplies the maps E_{i,j} to H'_{j,i} whose pullback and twist give the jointly surjective family onto F.

Source reference. Original proposition 01Q3: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9478-9483 to get the finite jointly surjective families on each U_j; the setup and use of the families is inside the counted span.

Source reference. Original lemma 01PG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9492-9496 to trim the pushforward modules; the application of the resolution property to the trimmed submodules is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes. Assume*

(1) *Y is quasi-compact and quasi-separated and has the resolution property,*
(2) *there exists an f-ample invertible module on X.*

Then X has the resolution property.

Proof. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $\mathcal{L}$ be an f-ample invertible module. Choose an affine open covering $Y = V_1 \cup \ldots \cup V_m$. Set $U_j = f^{-1}(V_j)$. By Properties, Proposition 01Q3 for each j we know there exists finitely

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many maps $s_{j,i} : \mathcal{L}^{\otimes n_{j,i}}|_{U_j} \to \mathcal{F}|_{U_j}$ which are jointly surjective. Consider the quasi-coherent $\mathcal{O}_Y$-modules

$$\mathcal{H}_n = f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$$

We may think of $s_{j,i}$ as a section over V_j of the sheaf $\mathcal{H}_{-n_{j,i}}$. Suppose we can find finite locally free $\mathcal{O}_Y$-modules $\mathcal{E}_{i,j}$ and maps $\mathcal{E}_{i,j} \to \mathcal{H}_{-n_{j,i}}$ such that $s_{j,i}$ is in the image. Then the corresponding maps

$$f^*\mathcal{E}_{i,j} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n_{i,j}} \longrightarrow \mathcal{F}$$

are going to be jointly surjective and the lemma is proved. By Properties, Lemma 01PG for each i, j we can find a finite type quasi-coherent submodule $\mathcal{H}'_{i,j} \subset \mathcal{H}_{-n_{j,i}}$ which contains the section $s_{i,j}$ over V_j. Thus the resolution property of Y produces surjections $\mathcal{E}_{i,j} \to \mathcal{H}'_{j,i}$ and we conclude. $\square$